

5-mavzu. To'la differensialli tenglamalar va unga keladigan tenglamalar (2-soat).

Reja:

1. To'la differensialli tenglama
2. Integrallovchi ko'paytuvchi va uni izlash usullari
3. Chiuiziqli differensial tenglama

To'la differensialli tenglama, integrallovchi ko'paytuvchi

To'la differensialli tenglama.

x va y o'zgaruvchilar teng huquqli qatnashgan differensial (differensiallarda yozilgan) tenglamani qaraylik:

$$M(x, y)dx + N(x, y)dy = 0 \quad (\{M, N\} \subset C(D)). \quad (1)$$

Agar D sohada (1) tenglamaning chap tomonidagi differensial ifoda **potensial** deb ataluvchi biror $u(x, y) \in C^1(D)$ funksiyaning to'la differensialidan iborat, ya'ni

$$M(x, y)dx + N(x, y)dy = 0, (M, N \in C(D)) \quad (2)$$

bo'lsa, u holda (1) tenglama D sohada **to'la (to'liq) differensialli tenglama** deyiladi.

To'la differensialli tenglama uchun differensial ta'rifi va (2) tenglikka ko'ra

$$\frac{\partial u}{\partial x} = M, \quad \frac{\partial u}{\partial y} = N \quad (3)$$

va, demak, bu tenglama $du(x, y) = 0$ ko'rinishga keladi. Bu holda, ravshanki, $y = \varphi(x) \in C^1(I)$ (yoki $x = \psi(y) \in C^1(I)$) funksiya (1) ning yechimi bo'lishi uchun I oraliqda $u(x, \varphi(x)) \equiv \text{const}$ (yoki $u(\psi(y), y) \equiv \text{const}$) ayniyat bajarilishi kerak.

Demak, oshkormas funksiya haqidagi teorema ko'ra to'la differensialli (1) tenglamaning har qanday regulyar nuqtasi atrofida integral chiziqlari ushbu

$$u(x, y) = c \quad (c - \text{const}) \quad (4)$$

tenglama bilan oshkormas ko'rinishda beriladi.

Bu yerda shuni e'tirof etaylikki, u potensial ixtiyoriy additiv o'zgarmas aniqligida topiladi, chunki ixtiyoriy o'zgarmas C uchun u bilan birgalikda $u + C$ ham potensial bo'ladi.

$Mdx + Ndy$ differensial ifoda biror funksiyaning to'la differensialidan iborat bo'lishi uchun shartlar matematik analiz kursida o'rganiladi. Biz bu yerda analizda isbotlanadigan quyidagi teoremani keltiramiz.

Teorema. Aytaylik, D soha bir bog'lamli, $\frac{\partial M}{\partial y}$ va $\frac{\partial N}{\partial x}$ hosilalar D da mavjud va uzluksiz bo'lsin. U holda (1) tenglamaning D to'la differensialli bo'lishi uchun D ning har bir nuqtasida

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (5)$$

shartning bajarilishi yetarli va zarurdir. Bu shart bajarilganda

$$u(x, y) = \int_{(x_0, y_0)}^{(x, y)} M(x, y)dx + N(x, y)dy \quad (6)$$

funksiyaning to'la differensial $Mdx + Ndy$ ifodadan iborat bo'ladi; bu yerdagi integral (x_0, y_0) va (x, y) nuqtalarni D da tutashtiruvchi ixtiyoriy yo'l bo'ylab olingan 2 – tur egri chiziqli integraldir ((6) integralning qiymati integrallash yo'lga bog'liq bo'lmaydi).

Shunday qilib, to'la differensialli tenglamani integrallash (6) formulaga ko'ra $u(x, y)$ potensialni topishga keltirildi, chunki to'la differensialli tenglamaning yechimi (4) formula bilan oshkormas ko'rinishda beriladi.

Misol 1. Ushbu

$$(x^2 + 2y)dx + (2x + y^2)dy = 0 \quad (7)$$

tenglama $D = \mathbb{R}^2$ tekislikda to'la differensiallidir, chunki bu holda

$$M = x^2 + 2y, N = 2x + y^2 \quad \text{va} \quad \frac{\partial M}{\partial y} = 2 = \frac{\partial N}{\partial x}$$

$u = u(x, y)$ potensialni topish uchun (3) shartlardan foydalanamiz;

$$u'_x = x^2 + 2y, u'_y = 2x + y^2;$$

$$u'_x = x^2 + 2y \Rightarrow u = \frac{x^3}{3} + 2xy + \varphi(y), u'_y = 2x + \varphi'(y);$$

$$u'_y = 2x + y^2 = 2x + \varphi'(y) \Rightarrow \varphi'(y) = y^2 \Rightarrow \varphi(y) = \frac{y^3}{3};$$

$$u = \frac{x^3}{3} + 2xy + \varphi(y) = \frac{x^3}{3} + 2xy + \frac{y^3}{3}.$$

Qaralayotgan misolda u funksiyani differensial xossalaridan foydalanib, bevosita topsa ham bo'ladi:

$$\begin{aligned} (x^2 + 2y)dx + (2x + y^2)dy &= x^2 dx + y^2 dy + 2y dx + 2x dy = \\ &= d\left(\frac{x^3}{3}\right) + d\left(\frac{y^3}{3}\right) + 2d(xy) = \\ &= d\left(\frac{x^3}{3} + \frac{y^3}{3} + 2xy\right). \end{aligned}$$

Demak, (7) tenglamaning yechimi

$$\frac{x^3}{3} + \frac{y^3}{3} + 2xy = \frac{c}{3}, c = \text{const},$$

ya'ni

$$x^3 + y^3 + 6xy = c$$

oshkormas ko'rinishda beriladi.

Integrallovchi ko'paytuvchi. «Agar berilgan tenglama uchun (5) shart bajarilmasa, tenglamani biror «integrallovchi» ko'paytuvchiga ko'paytirib, uni to'la differensialli ko'rinishga keltirish mumkinmi?» – degan savol tug'iladi.

Agar D sohada nolga aylanmaydigan $\mu = \mu(x, y) \in C^1(D)$ funksiya uchun

$$\mu M dx + \mu N dy = 0 \quad (8)$$

tenglama to'liq differensialli bo'lsa, u holda μ funksiya (I.8.1) tenglamaning **integrallovchi ko'paytuvchisi** deyiladi.

Faraz qilaylik, D – bir bog'lamli soha va $\{M, N\} \subset C^1(D)$ bo'lsin. (8) dan $\mu \in C^1(D)$ integrallovchi ko'paytuvchi uchun

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x} \quad (9)$$

shartni hosil qilamiz. (9) dan

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x} \quad (10)$$

Shunday qilib, μ integrallovchi ko'paytuvchi xususiy hosilali differensial tenglama (10) ning yechimi sifatida aniqlanishi kerak. Umumiy holda bu tenglamani yechish dastlabki tenglama (1) ni yechishdan oson emas. Lekin ba'zi hollarda (10) dan integrallovchi ko'paytuvchi μ ni topish uchun foydalanish mumkin.

Integrallovchi ko'paytuvchini biror $\omega = \omega(x, y)$ funksiyaning $\mu = \mu(\omega)$

(11)

funksiyasi sifatida izlab ko'raylik. (1.11)ni (1.10) ga qo'yamiz:

$$\left(M \frac{\partial \omega}{\partial y} - N \frac{\partial \omega}{\partial x} \right) \frac{d\mu}{d\omega} = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mu \quad (12)$$

Bu tenglik qanoatlanishi uchun

$$\frac{\frac{\partial N(x, y)}{\partial x} - \frac{\partial M(x, y)}{\partial y}}{M(x, y) \frac{\partial \omega(x, y)}{\partial y} - N(x, y) \frac{\partial \omega(x, y)}{\partial x}} \equiv f(\omega) \quad (13)$$

bo'lishi kerak, ya'ni (13) ning chap tomonidagi x va y ga bog'liq bo'lgan funksiya $\omega = \omega(x, y)$ ning funksiyasi sifatida ifodalanishi lozim. Bu holda (13) ni (12) ga qo'yib,

$$\mu = e^{\int f(\omega) d\omega} \quad (14)$$

integrallovchi ko'paytuvchini topamiz. (13) tenglik $\mu = \mu(\omega)$ shakldagi integrallovchi ko'paytuvchining mavjudlik shartini beradi. Mashqlar bajarganda ω funksiyaning $\omega(x, y) \equiv y$, $\omega(x, y) \equiv x$, $\omega(x, y) \equiv xy$, $\omega(x, y) \equiv x^2 + y^2$ va hokazo ko'rinishlarda tanlashga harakat qilib ko'rish mumkin.

Umumiy holda itegrallovchi ko'paytuvchining mavjud bo'lishi uchun yetarli va zaruriy shartlar Li gruppallari nazariyasida o'rganiladi.

Misol 2. Ushbu

$$3y dx + x(\ln y + 2 \ln x + 1) dy = 0 \quad (x > 0, y > 0)$$

differensial tenglamani yeching.

Bu tenglama to'la differensialli emas. $\mu = \mu(\omega)$, $\omega = \omega(x, y)$, ko'rinishdagi integrallovchi ko'paytuvchini topishga harakat qilamiz. Integrallovchi ko'paytuvchini

$$3y\mu dx + x(\ln y + 2\ln x + 1)\mu dy = 0$$

tenglamaning to'la differensialli bo'lishi shartidan, ya'ni

$$(3y\mu)'_y = (x(\ln y + 2\ln x + 1)\mu)'_x$$

tenglamadan topamiz. Zarur hisoblashlarni bajarib, oxirgi tenglamani quyidagi ko'rinishga keltiramiz:

$$(3y\omega'_y - x(\ln y + 2\ln x + 1)\omega'_x)\mu' = (\ln y + 2\ln x)\mu$$

Bu tenglamaning ko'rinishidan kelib chiqib, $\omega = \omega(x, y)$ funksiya uchun

$$\omega'_x = -\frac{1}{x} \quad \text{va} \quad 3y\omega'_y + 1 = 0$$

shartlarni qo'yamiz va $\mu = \mu(\omega)$ funksiya uchun $\mu' = \mu'$, ya'ni $\mu = e^\omega$ ekanligini topamiz. Endi zarur integrallashlarni va ixchamlashtirishlarni bajarib, integrallovchi ko'paytuvchini aniqlaymiz:

$$\omega'_x = -\frac{1}{x} \Rightarrow \omega = -\ln x + \psi(y), \quad \omega'_y = \psi'(y)$$

$$3y\omega'_y + 1 = 0 \Rightarrow \omega'_y = -\frac{1}{3y} = \psi'(y) \Rightarrow \psi(y) = -\frac{1}{3}\ln y$$

$$\omega = -\ln x + \psi(y) = -\ln x - \frac{1}{3}\ln y = \ln \frac{1}{xy^{1/3}}$$

$$\mu = e^\omega = \exp\left(\ln \frac{1}{xy^{1/3}}\right) = \frac{1}{xy^{1/3}}$$

Endi berilgan differensial tenglamani topilgan $\mu = \frac{1}{xy^{1/3}}$ integrallovchi ko'paytuvchiga ko'paytirib, ushbu

$$\frac{3}{x}y^{2/3}dx + \frac{1}{y^{1/3}}(\ln y + 2\ln x + 1)dy = 0$$

to'la differensialli tenglamani hosil qilamiz. Bu tenglamaning $u = u(x, y)$ potensiali uchun

$$u'_x = \frac{3}{x}y^{2/3}, \quad u'_y = \frac{1}{y^{1/3}}(\ln y + 2\ln x + 1)$$

bo'lishi kerak. Oxirgi ikki shartdan $u = u(x, y)$ funksiyani osongina topamiz:

$$u = 3y^{2/3}\ln x + \frac{3}{2}y^{2/3}\ln y - \frac{3}{4}y^{2/3}$$

Demak, berilgan tenglamaning yechimi

$$y^{2/3}(4\ln x + 2\ln y - 1) = c$$

tenglama bilan oshkormas ko'rinishda beriladi. □

Ba'zan (I.8.1) tenglamaning integrallovchi ko'paytuvchisini topish uchun chap tomonni ikkiga ajratib, tuzilgan tenglamalarning integrallovchi ko'paytuvchilarini topib, ular orqali izlangan integrallovchi ko'paytuvchini qurish mumkin bo'ladi. Aniqrog'i, quyidagicha ish tutish mumkin. (I.8.1) tenglamani

$$Mdx + Ndy \equiv (M_1dx + N_1dy) + (M_2dx + N_2dy) = 0$$

$$(M = M_1 + M_2, N = N_1 + N_2)$$

ko'rinishda yozib, $M_1dx + N_1dy = 0$ va $M_2dx + N_2dy = 0$ tenglamalar uchun μ_1

va μ_2 integrallovchi ko'paytuvchilarni topaylik; u holda

$$\mu_1 M_1 dx + \mu_1 N_1 dy = du_1 \text{ va } \mu_2 M_2 dx + \mu_2 N_2 dy = du_2$$

tengliklarga ega bo'lamiz. Ravshanki, ixtiyoriy noldan farqli uzluksiz φ_1 va φ_2 funksiyalar uchun

$$\varphi_1(u_1)\mu_1(M_1dx + N_1dy) = \varphi_1(u_1)(\mu_1 M_1 dx + \mu_1 N_1 dy) = \varphi_1(u_1)du_1 = d(\int \varphi_1(u_1)du_1)$$

$$\varphi_2(u_2)\mu_2(M_2dx + N_2dy) = d(\int \varphi_2(u_2)du_2)$$

bo'ladi. Endi, agar φ_1 va φ_2 funksiyalarni ushbu

$$\varphi_1(u_1)\mu_1 = \varphi_2(u_2)\mu_2$$

shartdan topsak, u holda ravshanki,

$$\begin{aligned} \varphi_1(u_1)\mu_1(Mdx + Ndy) &= \varphi_1(u_1)\mu_1(M_1dx + N_1dy) + \varphi_2(u_2)\mu_2(M_2dx + N_2dy) = \\ &= d(\int \varphi_1(u_1)du_1 + \int \varphi_2(u_2)du_2) \end{aligned}$$

bo'ladi. Demak, beilgan (1.1) tenglamaning integrallovchi ko'paytuvchisi

$$\mu = \varphi_1(u_1)\mu_1 = \varphi_2(u_2)\mu_2$$

funksiyadan iborat, yechimi esa

$$u \equiv \int \varphi_1(u_1)du_1 + \int \varphi_2(u_2)du_2 = c$$

munosabat bilan beriladi.

Mavzuni mustahkamlash uchun savollar:

1. To'la differensialli tenglama deb nimaga aytiladi?
2. Integrallovchi ko'paytuvchi va uni qanday izlash usullari bor?
3. To'la differensialli tenglamaning fizika fani bilan qanday bog'liqligi bor?